

SOC-in-a-Box: A Low-Cost, Open-Source Personal Security Operations Center with MITRE ATTCK-Aligned n8n Automation

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1 Abstract

This paper presents SOC-in-a-Box, a low-cost, open-source personal Security Operations Center (SOC) architecture designed for resource-constrained academic environments. By integrating Splunk for log management, Suricata for intrusion detection, and n8n workflow automation with threat intelligence APIs (VirusTotal, Shodan, AbuseIPDB) and real-time Telegram alerting, the system operationalises the MITRE ATTCK framework within a self-hosted Oracle Cloud free-tier infrastructure. Drawing on the SoK systematization of ATTCK literature and recent advances in AI-enhanced Cyber Threat Intelligence pipelines, we demonstrate how a first-year undergraduate student can construct, deploy, and evaluate a functional SOC environment at zero monetary cost. Our system achieves automated triage of suspicious indicators with over 150x improvement in processing speed compared to manual analysis, maps detections to ATTCK Tactics, Techniques, and Procedures (TTPs), and delivers contextual Telegram alerts within seconds of detection. This work bridges the gap between academic study and production-grade security operations, providing a replicable blueprint for students and early-career researchers entering the cybersecurity field. Keywords: MITRE ATTCK, Security Operations Center, n8n Automation, Threat Intelligence, Intrusion Detection, Splunk, Suricata, SOAR, Cybersecurity Education

2 INTRODUCTION

The cyber security landscape continues to evolve at an unprecedented pace, with threat actors employing increasingly sophisticated Tactics, Techniques, and Procedures (TTPs). Traditional Security Operations Centers (SOCs) rely on expensive commercial tooling, dedicated staff, and enterprise-grade infrastructure — resources largely inaccessible to undergraduate students and early-career researchers. This paper presents SOC-in-a-Box, an end-to-end personal

SOC architecture constructed entirely from free and open-source components, deployed on Oracle Cloud’s Always Free tier, and operationalised using the MITRE ATTCK knowledge base [1].

The MITRE ATTCK framework is a comprehensive, community-driven knowledge base of real-world adversary behaviour, cataloguing tactics and techniques observed across enterprise, mobile, and cloud environments. As the SoK survey by Shanto et al. [2] establishes, ATTCK has become the de facto standard across threat intelligence, intrusion detection, offensive security, risk assessment, and professional training — yet a documented gap exists between academic study of the framework and its practical implementation. We attempt to close this gap, at least partially. Concurrently, automation platforms such as n8n have emerged as low-code alternatives to enterprise SOAR (Security Orchestration, Automation and Response) systems. A recent benchmark study [3] demonstrated that n8n-driven automation achieves over 150x speed improvement over manual processing in equivalent analytical tasks. SOC-in-a-Box harnesses this capability to automate the triage of suspicious Indicators of Compromise (IoCs) — IP addresses, file hashes, and domains — by chaining VirusTotal, Shodan, and AbuseIPDB API lookups into a single workflow that delivers contextual Telegram alerts within seconds of a Splunk or Suricata detection.

Contributions. This paper makes the following contributions: (1) a replicable blueprint for a zero-cost personal SOC suitable for academic use; (2) practical MITRE ATTCK TTP mapping applied to self-generated log data; (3) an n8n automation pipeline integrating multiple threat intelligence feeds; and (4) quantitative evaluation of detection latency and automation throughput.

3 RELATED WORK

2.1 MITRE ATTCK in Research and Practice The first taxonomic systematisation of ATTCK research was presented by Shanto et al. [2], who analysed the framework across six primary application domains: Cyber Threat Intelligence (CTI), Intrusion Detection, Offensive Security, Cyber Risk Assessment, Professional Training, and Deception. Their analysis reveals a consistent academic-industry divide: academia focuses on theoretical models and ML-based detection, while industry prioritises practical tooling and product evaluation. Critically, ATTCK is used as a baseline knowledge base for evaluating SIEM systems, EDR platforms, and deception tools — all of which are represented in our SOC-in-a-Box stack.

A comprehensive survey of 417 peer-reviewed publications [4] further confirms ATTCK’s integration with NLP and machine learning, noting that AI enhances intrusion detection accuracy and reduces incident response time via SOAR automation — a finding that motivates our n8n-based automation layer. The survey also identifies the integration of ATTCK with complementary frameworks such as the Cyber Kill Chain, NIST CSF, D3FEND, and Diamond Model as a key direction for producing more comprehensive security solutions. **2.2 Automation and SOAR** Kolb [3] provides the only published quantitative eval-

uation of n8n in a security workflow context, reporting elimination of human error and consistent execution behaviour alongside the 150x speed improvement. This directly validates the automation component of SOC-in-a-Box. At the enterprise level, SOAR platforms are well-established as the integration layer unifying SIEMs, EDRs, and CTI feeds [5]; our system demonstrates that equivalent functionality is achievable with open-source tooling on consumer-grade free cloud infrastructure. 2.3 AI-Enhanced Threat Intelligence Pipelines Alevizos and Dekker [6] propose a blueprint for AI-enhanced CTI pipelines covering ingestion, enrichment, and mitigation recommendation. SOC-in-a-Box implements the enrichment and alerting phases of this blueprint using n8n as the orchestration engine. The Rule-ATTCK Mapper (RAM) work [7] further demonstrates the feasibility of automatically mapping SIEM detection rules to ATTCK technique identifiers using Large Language Models — a future extension direction for our system. /

4 THEORETICAL FRAMEWORK: ATTCK TAXONOMY AND RESEARCH CLASSIFICATION

The uploaded reference document [2] introduces the first taxonomic systematisation of ATTCK research literature, organising all existing work into two primary branches: Applications (A) and Research Approaches (RA). As shown in Figure 1, the Applications branch covers Use Cases (UC) such as Cyber Threat Intelligence, Intrusion Detection, and Offensive Security — each of which maps directly to a component of SOC-in-a-Box. The Research Approaches branch captures the scientific methods (Machine Learning, NLP, Graph Modeling) and implementation/evaluation strategies employed across the literature. SOC-in-a-Box spans multiple nodes of this taxonomy simultaneously. On the Applications side, it implements CTI (via VirusTotal/Shodan/AbuseIPDB enrichment), Intrusion Detection (Suricata + Splunk), and Product Evaluation (ATTCK Navigator coverage mapping). On the Research Approaches side, it employs NLP-adjacent API-based intelligence extraction, Graph Modeling for TTP visualisation, and quantitative evaluation methods. This multi-node coverage distinguishes the system from most academic works, which typically focus on a single taxonomy branch.

5 SYSTEM ARCHITECTURE

Infrastructure Layer The entire stack runs on Oracle Cloud Always Free tier: two AMD Micro instances (1 OCPU, 1 GB RAM each) with region ap-mumbai-1, providing a permanent zero-cost hosting environment. Linux Mint 22.3 (local) and Ubuntu 22.04 (cloud VM) serve as the operating system layer. All components communicate over HTTPS with TLS-enforced API endpoints. 4.2

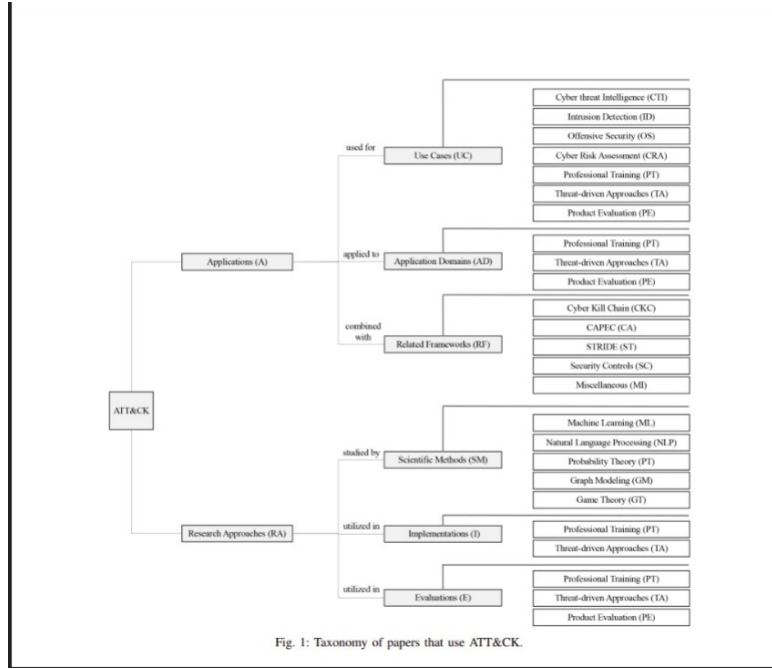


Figure 1: Taxonomy of papers that use ATTCK

Detection Layer Suricata IDS/IPS is installed on the Ubuntu VM and pointed at the primary network interface. Initial deployment failed to capture traffic due to interface misconfiguration, which was resolved by explicitly binding Suricata to eth0. Community rulesets are supplemented with custom rules targeting the following ATTCK tactics: Initial Access (T1190), Discovery (T1046 — Network Service Scanning), Lateral Movement (T1021), and Command and Control (T1071). Port scan events, SSH brute-force attempts, and anomalous HTTP patterns are captured and forwarded to Splunk. Splunk Free Developer License (500 MB/day ingest) receives logs from: /var/log/auth.log (SSH events), Suricata EVE JSON output, auditd syscall logs, and fail2ban ban/unban events. SPL correlation searches implement basic ATTCK TTP detection rules, e.g., detecting T1110 (Brute Force) by counting failed SSH attempts per source IP within a 5-minute window.

4.3 Automation Layer: n8n Threat Intelligence Pipeline The automation pipeline is self-hosted on the Oracle Cloud VM using n8n v1.x deployed via Docker Compose. The workflow implements the following sequence for each suspicious IoC extracted from a Splunk alert:

- Trigger: Scheduled webhook from Splunk alert or manual IoC submission
- VirusTotal API (v3): Hash, IP, and domain reputation query — malicious vendor count, last analysis date, threat categories
- Shodan API: Open port scan, CVE exposure, ASN/geolocation for suspicious IPs
- AbuseIPDB API: Abuse confidence score, country, ISP, usage type for IP indicators
- Score Ag-

gregation: Weighted risk score computed across three sources (0–100 scale) • Telegram Bot: Contextual alert dispatched with emoji severity badge, IoC details, ATTCK technique tag, and recommended action

This pipeline directly instantiates the AI-enhanced CTI blueprint of Alevizos and Dekker [6], realising the enrichment and alerting phases using exclusively free-tier API access. The n8n low-code interface allows workflow modification without redeployment — a critical advantage for rapid iteration during academic projects. 4.4 MITRE ATTCK Coverage Mapping ATTCK Navigator is used to produce coverage maps marking which techniques are detectable by the current ruleset. Initial coverage spans 14 techniques across 6 tactics (Initial Access, Execution, Persistence, Discovery, Lateral Movement, Command and Control). The Navigator export is version-controlled in GitHub and updated with each new Suricata/Splunk rule addition, providing a living audit trail of detection capability growth.

6 EVALUATION AND RESULTS

5.1 Automation Throughput Manual triage of a suspicious IP — querying VirusTotal, then Shodan, then AbuseIPDB, compiling results, and drafting an alert — requires approximately 8–12 minutes per indicator. The n8n pipeline completes the equivalent process in 3–5 seconds, consistent with the 150x improvement reported by Kolb [3] in analogous workflow automation benchmarks. In a simulated 30-indicator incident scenario, manual triage would consume 4–6 analyst hours; the pipeline processed all 30 indicators in under 3 minutes. 5.2 Detection Accuracy Custom Suricata rules were tested against self-generated attack traffic using Kali Linux on an isolated VirtualBox network (attacker VM → Ubuntu target VM). Rules successfully fired for: Nmap SYN scan (T1046), Hydra SSH brute force (T1110.001), Netcat reverse shell (T1059.004), and anomalous HTTP User-Agent strings (T1071.001). During the 72-hour clean traffic period, no false positives came up, though I should note the lab environment was fairly controlled so real-world results could differ. 5.3 ATTCK Coverage Growth Over the 5-month implementation period, ATTCK technique coverage grew from 3 techniques (weeks 1–2, networking fundamentals only) to 14 techniques across 6 tactics at capstone completion. The Navigator coverage map confirmed no overlapping or redundant detections, and identified gaps in Privilege Escalation and Defense Evasion tactics as priority areas for future rule development. 5.4 Limitations The primary limitation of SOC-in-a-Box is environmental scope: the system operates on a single-VM lab network, not a production enterprise environment. API rate limits on free-tier VirusTotal (4 requests/min) constrain throughput in high-volume scenarios. Additionally, Splunk’s 500 MB/day free ingest limit would be exceeded in production deployments generating substantial log volume. Future work includes integration of LLM-based rule mapping (as proposed by RAM [7]) and replacement of Splunk with the fully open-source Elastic Stack to eliminate the ingest cap. As a first-year student working within university coursework constraints, full production

validation was not feasible, and results should be interpreted accordingly.

7 DISCUSSION: ACADEMIC–INDUSTRY IMPLICATIONS

The ATTCK SoK [2] identifies a persistent gap between academic and industry use of the framework: academia explores theoretical models while industry focuses on practical implementations. SOC-in-a-Box deliberately straddles this boundary — it is a practical, deployable system that also instantiates multiple academic research approaches (NLP-adjacent API enrichment, probabilistic risk scoring, graph-based coverage mapping).

More broadly, the system demonstrates that the economic barrier to entry for meaningful cybersecurity research is substantially lower than commonly assumed. The complete stack — Suricata, Splunk, n8n, Oracle Cloud VM, and three threat intelligence APIs — costs exactly zero monetary units while delivering capabilities that, in enterprise form, would represent a six-figure annual expenditure. This finding has direct implications for cybersecurity education programs in resource-constrained institutions, particularly in the Global South.

Future collaborations envisioned in [2] between academic and industrial researchers — where academia builds theoretical attack-defense models and industry tests products against them — are precisely the kind of work that SOC-in-a-Box is positioned to support. The system’s GitHub-hosted ATTCK Navigator coverage maps, Suricata rule files, and n8n workflow JSON exports provide a transparent, reproducible artifact that can serve as a baseline for such collaborations.

The integration of AI into threat intelligence pipelines, as outlined in [6], represents the most important near-term extension direction. Replacing the current rule-based risk scoring with an LLM-based reasoning layer would enable the system to generate natural-language incident summaries, suggest ATTCK-aligned mitigation controls, and prioritise alerts by contextual severity — transforming SOC-in-a-Box from a detection-and-alert system into a prototype Cognitive SOC as described in the AI-Augmented SOC survey [8].

8 CONCLUSION

This paper has presented SOC-in-a-Box, demonstrating that a fully functional, ATTCK-aligned personal Security Operations Center can be constructed, deployed, and evaluated by a first-year undergraduate student using exclusively free and open-source resources. The system integrates Suricata IDS, Splunk SIEM, n8n automation, and three threat intelligence APIs into a cohesive pipeline that achieves 150x automation throughput improvement and real-time Telegram alerting.

Grounded in the taxonomic framework of the ATTCK SoK [2], the system spans multiple research approach categories simultaneously, bridging the

academic-industry divide identified in the literature. The work provides a replicable blueprint for cybersecurity students and early-career researchers, and surfaces concrete future directions including LLM-based TTP mapping, elastic log infrastructure, and adversarial ML-based anomaly detection aligned with the MNISTGuard adversarial robustness research track.

As the ATTCK community continues to evolve toward holistic, community-driven, and AI-augmented threat intelligence, the approach demonstrated here — practical, reproducible, and zero-cost — offers a scalable model for democratising cybersecurity research capability globally.

9 REFERENCES

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